

Pradyumna “Prady” Manur

SKILLS

Engineering & Design: CAD (Autodesk Inventor), OpenRocket, 3D Printing, Hardware Prototyping

Programming Languages: Python

Technical Exposure & Learning: CFD, C++, VS Code PlatformIO, Microelectronics, Avionics Stack Integration, Sensor Data Processing, Flight Dynamics & Control Loop Simulation (PID Tuning)

Leadership & Execution: Technical Mentorship, Project Planning, Cross-Functional Communication, Team Collaboration

EXPERIENCE

Eagle Scout, Scouting America | Mar. 2019 – Jun. 2026

- Planned and executed an Eagle Scout Service Project, leading a team of 25 Scouts and adults to construct a 110-cubic-foot wooden composter benefiting Evanston Grows, a local urban farming organization that counters food deserts by distributing low-to-no-cost fresh produce to underserved communities.
- Served in multiple elected and appointed leadership roles, including Patrol Leader (Dragon Patrol), Assistant Patrol Leader, Troop Webmaster, Outdoor Ethics Guide, and Librarian.
- Completed intensive leadership training at the National Youth Leadership Training (NYLT) camp.
- Managed troop logistics and operations, leading and instructing a five-Scout patrol to conduct formal Troop training curricula and field activities.
- Inducted as an Order of the Arrow Ordeal Member, recognizing peer-elected service and leadership.
- Contributed to community development through annual popcorn fundraising campaigns and diverse service events, including trail conservation, infrastructure building (flagpoles, ax yards), and meal packing for Feed My Starving Children.

Co-Founder and Chief Engineer, E-Town Rocket Bureau | Aug. 2024 – Jun. 2026

- Co-founded E-Town Rocket Bureau and established its technical framework, leading engineering efforts to build high-precision model rockets for the 2025 and 2026 American Rocketry Challenge (ARC).
- Developed ATOS, a Python-based SITL flight simulator suite in Google Colab that automates PID control loop tuning via binary search algorithms, featuring an LLM-generated web interface for rapid testing.
- Engineered an ESP32 avionics stack and active-control airbrake system governed by a real-time PID control loop to optimize target apogee.
- Utilized Autodesk Inventor and Autodesk CFD alongside 3D printing and laser cutting to design, simulate, and rapidly prototype aerodynamic flight hardware.
- Designed and deployed a central team website built using LLMs; created a public project showcase and digital repository to preserve CAD models, C++/Python codebases, and technical documentation for institutional memory.
- Mentored a cohort of 5+ teammates in Autodesk Inventor CAD modeling, flight physics, embedded systems, and practical aerospace engineering concepts.
- Drove 75% YoY recruitment growth via club fairs, successfully rebuilding team capacity after losing half the membership to graduation.

Team Member, North Suburban Math League | Sep. 2022 – May 2026

- Competed in regional mathematics contests, executing analytical problem-solving under constrictive timeframes.
- Collaborated with peers to develop out-of-the-box problem-solving strategies.

Intern, Ward Manufacturing Co. | Mar. and Jul. 2025

- Operated a Haas GT-20 CNC Lathe to precision-manufacture over 4,350 components, including 4,241 specialized components destined for aerospace systems.
- Spearheaded a Quality Control inventory overhaul, locating, identifying, tagging, photographing, and digitizing physical QC gages.
- Assisted the quality control department in transferring legacy inventory systems into 1Factory software.
- Deconstructed decommissioned tool and die setups to isolate, evaluate, and salvage reusable precision components.
- Gained cross-functional exposure to CNC software programming, tool room maintenance, manufacturing production workflows, industrial marketing, and facility management operations.

EDUCATION & HONORS

- Awards & Recognition:
 - Michael Wynn Circle for Dedication, Inspiration, and Compassion (2026)
 - Justin Wynn Award for Leadership, Citizenship, and Sportsmanship (2018)
- Eagle Scout (December 2025 onwards)
- Evanston Township High School | Highest Honor Distinction (2026)
- Standardized Testing: 1550 SAT (Spring 2025: 780 Math, 770 Reading & Writing)